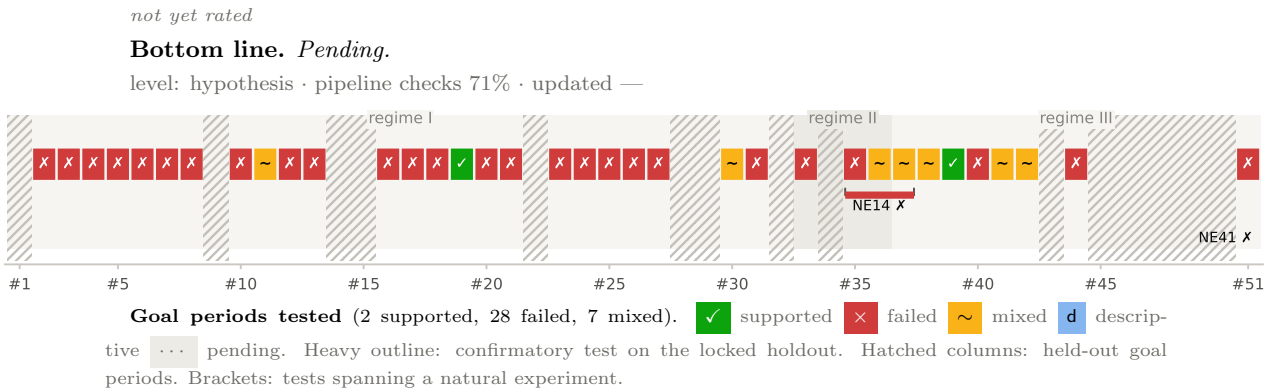


H42: Cross-excitation is a delayed step at the next read-out

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04



WHAT HAPPENED

Not yet summarized. The question: Does a multivariate Hawkes model whose cross-excitation fires only at the recipient’s next model call fit better than exponential kernels, raise the cross-branching estimate, and make it consistent across regimes? Why it should. H03 found swarm activity sub-critical ($n \approx 0.2\text{--}0.4$) with only fast cross-triggering ($n_x \approx 0.07$, $\tau \approx 10\text{--}30$ s) beating an agent-shift null. H08 showed responses wait for the recipient’s next model call (read-out gating). H25 showed the equal-time dial is blind to coupling delayed by minutes. The ledger’s read-out lag (message $\rightarrow$ recipient’s first call that sees it) has median ≈ 20 s but a long tail ($q_{75} \approx 75$ s, $q_{90} \approx 150\text{--}200$ s, $q_{99} \approx 6\text{--}16$ min in #27/#38: pauses, long to

PHYSICAL MODEL

From: physics-models/09-hawkes/ (multivariate, fixed-timescale kernel grids, H03’s M3 parameterization: homogeneous self and cross weights, agent-day baselines). Each realization is one (unit, village day); t is seconds since the day’s window start (clipped to the unit’s start/end where a day is split). Excitation never carries across days. Agent i is at risk only inside its span (its first call start to its last cal

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.
Observables. *Pending.*

RESULTS VS PREDICTIONS

Pending.

GOAL PERIODS IN DETAIL

G02 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G03 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G04 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G05 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G06 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G07 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G08 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G10 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G11 ~ mixed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G12 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G13 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G16 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G17 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 G18 × failed (amended reading A1/A3: world-B read-out gain vs world-A exp...
 +23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	–	F identifiability	–
B assumptions	–	G ground truth	–
C adequacy	–	H comparative	–
D unfitted predictions	–	I transfer	–
E interventional	–		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pending.

ROUND 2

No round-2 redirects yet.

H42-readout-hawkes-kernel : generated 2026-10-04 01:33 by infra/summaries/build-summaries.py : card hypotheses/H42-readout-hawkes-kernel/README.md : ratings: not rated : DRAFT: auto stub, not yet written (page 2)